Face-Tail Diagnostics for Constraint Injection under Legendre p-Coarsening

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Abstract

Reducing the degree of a spectral element can violate a first-order reduction constraint even when the source representation satisfies it. For Legendre modal restriction, we express this transfer defect exactly through two endpoint values of the discarded field. The resulting rank-at-most-two commutator gives an explicit sharp tail-to-constraint amplification factor, an exact treatment of inherited constraint error, and a low-cost test of a proposed coarsening. We also quantify the field-error cost of the classical endpoint-preserving H^1 projector and identify a coefficient obstruction to repairing the potential while retaining the independently restricted derivative. Tensor-product extensions use face functions rather than two global scalars; nonaffine maps add a separate multiplication commutator. Reproducible checks include exact rational integration, 3,000 random polynomial transfers, 495 weighted singular-value comparisons, 2,048 gauge-wave metric-component transfers, and periodic discontinuous Galerkin scalar-wave evolutions. In the gauge-wave test, a field-only threshold accepts 1,360 candidates, of which 224 violate an independently specified constraint threshold; the exact joint test rejects those 224. Compatible projection removes the immediate injection but does not consistently reduce later evolution error. The contribution is an explicit diagnostic and transfer-cost analysis for spectral adaptation, with direct relevance to first-order numerical-relativity systems, rather than a new constraint-projection method or an Einstein evolution scheme.

Keywords: numerical relativity; spectral adaptation; reduction constraints; Legendre projection; discontinuous Galerkin methods; reproducible computation.

Code and data: https://github.com/shi1720/legendre-face-tail

1 Introduction

In a first-order evolution system, a field and its spatial derivative can be independent numerical variables. For a scalar wave these are a potential ψ and Φ_i, constrained by $C_i = \partial_i\psi - \Phi_i = 0$. The generalized harmonic formulation has the analogous metric reduction constraint $C_{iab} = \partial_i g_{ab} - \Phi_{iab}$ [1, 2]. A resolution change acts on both sides of this relation. Restricting the variables separately need not give the same result as differentiating the restricted potential.

This issue is relevant to adaptive spectral simulations even before the time integrator or boundary conditions enter. Spectral adaptive mesh refinement already monitors truncation, unresolved modes, and constraints [3]. SpECTRE supports local polynomial-degree changes within a task-based discontinuous Galerkin (DG) framework [4] that has enabled binary black hole evolutions [5]. A useful adaptation diagnostic should distinguish the constraint present before a transfer from the constraint created by the transfer itself.

There is a particularly concrete implementation question. At the inspected SpECTRE source revision, the constraint-based AMR criterion uses a coarsening factor and explicitly suggests evaluating constraints after a lower-order projection if choosing that factor proves difficult [6]. Here we analyze that prospective calculation for full-element Legendre restriction. The question of inspecting a candidate coarse state is therefore not new. Our purpose is to expose its exact structure, its worst-case amplification, and the limitations of a compatible alternative.

The central result is that the injected reduction constraint depends on only two discarded endpoint traces in one dimension. Small discarded L^2 energy does not determine these traces: equal-energy tails can inject zero constraint or attain the sharp worst case. Once modal coefficients are available, parity sums recover the complete new constraint without constructing a coarse nodal state. The same sums also give the exact extra field-error cost of a familiar endpoint-preserving H^1 projection.

The paper makes three connected contributions. First, it gives a closed-form transfer identity, its sharp finite-degree norm, and an exact separation of inherited and newly created constraint error. Second, it relates this identity to modal retention, endpoint preservation, and the classical compatible projector, including an explicit repair-cost formula. Third, it tests the diagnostic with independent algebra, nodal transfers, a metric-component benchmark, and a time-dependent DG solver. The experiments distinguish an instantaneous transfer guarantee from a claim about later waveform accuracy.

2 Relation to established work

Constraint projection and constraint-preserving boundary conditions have a substantial numerical-relativity literature. Holst et al. construct optimal projections using a field-space metric and demonstrate them for scalar fields on a black hole background [1]. Lindblom et al. develop the first-order generalized harmonic system with damping and boundary conditions [2]; Rinne studies stable radiation-controlling boundary conditions [7]. These results concern the constrained evolution problem and its boundary treatment. We study the algebra of a particular change of representation. A local transfer diagnostic cannot replace those methods.

Commuting projections are also classical. Projection-based interpolation [8] and finite element exterior calculus [9] organize approximation spaces so that projection respects differential relations. The endpoint-preserving one-dimensional H^1 projection used below is an instance of this established approach. Neither its construction nor its derivative-minimizing property is claimed as new.

Noncommutation of orthogonal polynomial truncation and differentiation is explicitly treated by Rapisarda, van Waarde, and Çamlibel [10], including the parity-structured Legendre differentiation matrix. Thus the existence of a commutator is not a novelty claim either. We specialize that structure to reduction-constraint transfer and derive the two-trace norm, sharp finite-tail gain, exact repair penalty, and candidate decision test in a common notation. These are elementary consequences of classical polynomial identities, supplied with proofs and independently reproducible numerical checks.

Finally, Szilágyi's projected constraint error [3, Sec. 6.1.4] projects derivative indices into grid-topology directions. That coordinate projection differs from restricting polynomial degree. His work also shows that removing a constraint-violating pile-up mode can *reduce* constraints. Our inherited-error formula permits that behavior: the inherited error and the new commutator can cancel. The diagnostic does not assert that coarsening always makes constraints larger.

3 Transfer setting and exact face-tail identity

Let $I = [-1, 1]$, let P_n be the standard Legendre polynomial normalized by $P_n(1) = 1$, and write

$$u(\xi) = \sum_{n=0}^{N} a_n P_n(\xi), \qquad \mathcal{T}_q u = \sum_{n=0}^{q} a_n P_n(\xi), \qquad 0 \le q < N. \tag{1}$$

Here $\mathcal{T}_q$ is the continuous unweighted $L^2(I)$ projector, and $D = \partial_\xi$. The normalization is

$$\|u\|^2 = \sum_{n=0}^{N} \frac{2a_n^2}{2n+1}. \tag{2}$$

All fields are real; the complex case follows with conjugation. An affine physical element of length h has $\partial_x = (2/h)D$.

For a source pair (u, v) with logical reduction constraint $C = Du - v$, componentwise restriction gives

$$C_q = D\mathcal{T}_q u - \mathcal{T}_q v = \mathcal{T}_q C + c_q(u), \qquad c_q(u) := (D\mathcal{T}_q - \mathcal{T}_q D)u. \tag{3}$$

The first term is inherited constraint error. The second is introduced solely by restriction. This decomposition is exact and does not assume $C = 0$.

Define the discarded parity sums and endpoint traces by

$$s_p = \sum_{\substack{q<n\le N \\ n\equiv p \ (2)}} a_n, \quad p \in \{0, 1\}, \qquad r = u - \mathcal{T}_q u, \qquad e_\pm = r(\pm 1). \tag{4}$$

Then $s_0 = (e_+ + e_-)/2$ and $s_1 = (e_+ - e_-)/2$.

Theorem 1 (Two-trace transfer identity)**.** *For the transfer in* (1)*,*

$$c_q(u) = -\sum_{k=0}^{q} (2k+1) s_{1-(k \bmod 2)} P_k, \tag{5}$$

$$\|c_q(u)\|^2 = 2\sum_{k=0}^{q} (2k+1) s_{1-(k \bmod 2)}^2. \tag{6}$$

The discarded-tail-to-constraint map has rank at most two. For $q \ge 1$ and $N \ge q+2$, its rank is exactly two.

Proof. Since $D\mathcal{T}_q u \in \mathbb{P}_{q-1}$, $c_q = -\mathcal{T}_q Dr$. For $k \le q$, integration by parts and Legendre orthogonality give

$$\int_{-1}^{1} Dr\, P_k \,\mathrm{d}\xi = e_+ - (-1)^k e_- - \int_{-1}^{1} r\, DP_k \,\mathrm{d}\xi = e_+ - (-1)^k e_-. \tag{7}$$

The last integral vanishes because $DP_k \in \mathbb{P}_{q-1}$. Multiplication by $(2k+1)/2$ gives the coefficient of $\mathcal{T}_q Dr$, which proves (5). Orthogonality gives (6). Even and odd discarded modes excite disjoint retained parities. Both are nonempty under the stated degree conditions, proving rank two. □

The proof uses only the residual's orthogonality and its endpoint traces. The identity therefore extends to $u \in H^1(I)$ with $r = u - \mathcal{T}_q u$, without requiring a convergent endpoint evaluation of an infinite coefficient series. The finite polynomial setting remains the computational setting of this paper. For $q = 0$, or for a single discarded mode, the rank can be one.

Corollary 2 (Exact prospective constraint test)**.** *On an affine one-dimensional element, independently restricting u and a physical derivative variable Φ gives*

$$C_{x,q} = \mathcal{T}_q C_x + \frac{2}{h} c_q(u), \qquad C_x = \frac{2}{h} Du - \Phi. \tag{8}$$

Its physical squared norm is $(h/2)\,\|C_{x,q}\|^2$. Testing this quantity against a specified squared tolerance is equivalent, in exact arithmetic, to forming the coarse fields and differentiating them.

Adding $\|\mathcal{T}_q C_x\|$ and $(2/h)\,\|c_q\|$ gives a sufficient bound, but loses possible cancellation. The implementation retains the signed coefficients in (8). Monitoring only the pre-transfer constraint cannot detect the injection from an initially constraint-satisfying state.

4 Sharp amplification and the cost of compatible transfer

4.1 A sharp finite-degree bound

Let

$$A_{q,p} = 2 \sum_{\substack{0 \le k \le q \\ k \not\equiv p \ (2)}} (2k+1), \qquad B_{Nq,p} = \frac{1}{2} \sum_{\substack{q < n \le N \\ n \equiv p \ (2)}} (2n+1). \tag{9}$$

An empty sum is zero.

Theorem 3 (Sharp tail amplification)**.** *The exact operator norm from discarded field L^2 norm to injected constraint L^2 norm is*

$$G_{Nq} = \max_{p \in \{0,1\}} \sqrt{A_{q,p} B_{Nq,p}}, \qquad \|c_q(u)\| \le G_{Nq}\, \|u - \mathcal{T}_q u\|\,. \tag{10}$$

Equality is attained by a tail supported on a maximizing parity with a_n proportional to $(2n+1)/2$ on that parity.

Proof. Set $E_p = \sum_{n>q,\ n\equiv p} 2a_n^2/(2n+1)$. Weighted Cauchy–Schwarz gives $s_p^2 \le B_{Nq,p} E_p$, with equality for the stated coefficients. Equation (6) becomes $\|c_q\|^2 = \sum_p A_{q,p} s_p^2$. Bounding each product by $\max_p A_{q,p} B_{Nq,p}$ times E_p, then summing, proves the bound. Concentrating the tail energy on a maximizing parity attains it. □

For fixed q, the gain grows linearly with N. An L^2-only infinite-tail assumption cannot bound the transfer constraint uniformly. For a physical element the ratio between physical constraint norm and physical discarded-field norm is $(2/h)G_{Nq}$. This explains why the same coefficient tail can become more consequential on smaller elements.

The gain is a worst-case statement, not a typical-case prediction. At $(N, q) = (24, 8)$, a unit-norm maximizing tail injects $108.995413\ldots$, whereas a normalized multiple of $P_9 - P_{11}$ injects zero. The latter tail has zero traces at both endpoints. A tail-energy threshold discards the information that distinguishes these cases (Fig. 1).

4.2 Classical compatible projection in two-mode form

For $q \ge 1$, let m_p be the largest integer at most q of parity p, and define

$$H_q u = \mathcal{T}_q u + s_0 P_{m_0} + s_1 P_{m_1}. \tag{11}$$

This is the endpoint-preserving H^1 projector, expressed through the same two tail sums. It is a classical compatible approximation operator; the following properties clarify what it changes in this transfer setting.

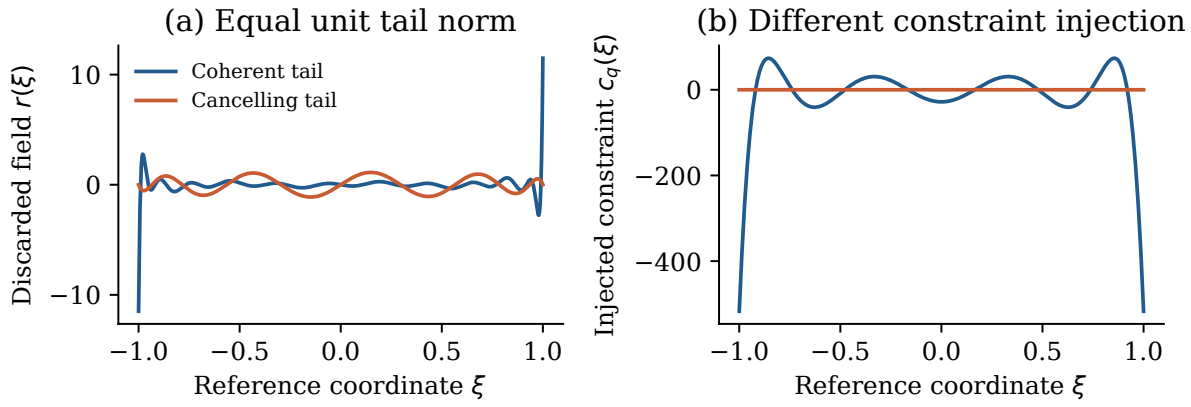


Figure 1: Two discarded tails at $N = 24, q = 8$, each with unit continuous L^2 norm. The coherent tail attains the sharp gain; the cancelling tail has zero endpoint traces and injects no logical reduction constraint.

Proposition 4 (Compatibility, retained modes, and field penalty)**.** *For $q \geq 1$,*

$$(H_q u)(\pm 1) = u(\pm 1), \qquad DH_q u = \mathcal{T}_{q-1} Du, \tag{12}$$

$$\|u - H_q u\|^2 = \|u - \mathcal{T}_q u\|^2 + \frac{2s_0^2}{2m_0+1} + \frac{2s_1^2}{2m_1+1}. \tag{13}$$

It minimizes $\|D(u-w)\|$ among $w \in \mathbb{P}_q$ with either endpoint anchored to u, and $\|DH_q u\| \leq \|Du\|$. For $q \geq 2$, it uniquely preserves both endpoint values and every Legendre coefficient through degree $q-2$. For $q = 1$, its constant coefficient can change.

Proof. At the endpoints, $P_{m_0}(1) = P_{m_0}(-1) = 1$ and $P_{m_1}(1) = -P_{m_1}(-1) = 1$, so the correction restores $e_\pm$. The classical derivative identity $DP_n = \sum_{k<n,\ n-k \text{ odd}} (2k+1)P_k$ [10, 11] shows that differentiating the two correction modes adds precisely the tail contributions of Du through degree $q-1$. Hence $DH_q u = \mathcal{T}_{q-1} Du$. Equation (13) follows because the discarded tail and the two retained correction modes are mutually orthogonal. Every polynomial in $\mathbb{P}_{q-1}$ is the derivative of a unique endpoint-anchored polynomial in $\mathbb{P}_q$. Orthogonal projection of Du therefore proves minimality and contraction. Finally, with coefficients through $q-2$ fixed, the two endpoint equations uniquely determine the last two coefficients and give (11). □

For a consistent scalar-wave state, the transfer

$$u \mapsto H_q u, \qquad \Pi \mapsto \mathcal{T}_q \Pi, \qquad \Phi \mapsto \mathcal{T}_{q-1} \Phi \tag{14}$$

preserves the reduction constraint on an affine element. It contracts the flat-space wave energy $\frac{1}{2}\int(\Pi^2+\Phi^2)\,\mathrm{d}x$, since each derivative variable is orthogonally projected. This statement concerns consistent source data and this particular energy. It does not apply to a general relativistic symmetrizer or imply contraction of an energy containing an independent u^2 term.

For inconsistent data the new logical constraint is $\mathcal{T}_{q-1}(Du - v)$, not necessarily zero. Endpoint preservation applies to u; it does not preserve characteristic combinations involving Π and Φ. These distinctions matter at DG interfaces.

Proposition 5 (Obstruction to potential-only repair)**.** *There exists $w \in \mathbb{P}_q$ satisfying $Dw = \mathcal{T}_q Du$ if and only if $s_{1-(q \bmod 2)} = 0$. When this condition holds, anchoring one endpoint gives $w = H_q u$, which also preserves the other endpoint.*

Proof. The coefficient of P_q in $\mathcal{T}_q Du$ is $(2q+1)s_{1-(q \bmod 2)}$, whereas Dw has degree at most $q-1$. Its vanishing is necessary and sufficient. When it vanishes, $\mathcal{T}_q Du = \mathcal{T}_{q-1} Du$, so Proposition 4 supplies the unique endpoint-anchored primitive. □

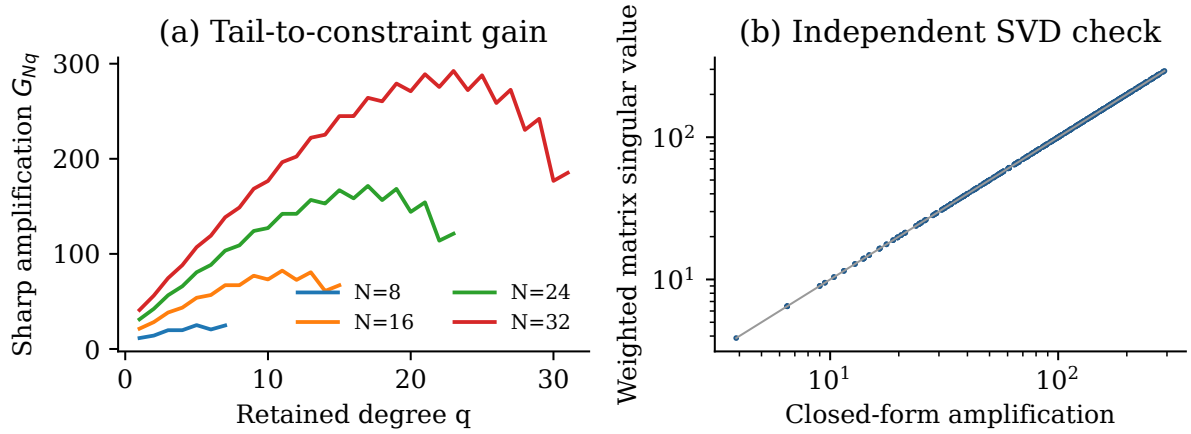


Figure 2: Sharp finite-tail gain. Left: its dependence on source and retained degrees. Right: closed-form and independently assembled weighted-matrix singular values for all 495 pairs $3 \le N \le 32, 1 \le q < N$. The largest relative difference is 7.1×10^{-16}.

Thus changing only the potential cannot generically repair the constraint while retaining all independently restricted derivative coefficients. Setting $\Phi = D\mathcal{T}_q u$ repairs the algebraic constraint instead, but alters derivative information and need not contract wave energy. Neither repair is universally preferable to rejecting the coarsening.

5 Tensor products, mappings, and implementation

5.1 Face functions on a tensor-product element

Let $u = \sum_{\boldsymbol{n}} a_{\boldsymbol{n}} \prod_j P_{n_j}(\xi_j)$, with degree caps N_j, and let $\mathcal{T}_{\boldsymbol{q}} = \prod_j \mathcal{T}_{q_j}^{(j)}$. Fix direction i. For every retained transverse multi-index $\boldsymbol{\ell}$, define

$$s_{i,p,\boldsymbol{\ell}} = \sum_{\substack{q_i < n_i \le N_i \\ n_i \equiv p \ (2)}} a_{(\boldsymbol{\ell}, n_i)}. \tag{15}$$

Here $(\boldsymbol{\ell}, n_i)$ inserts n_i in slot i. Because transverse restriction commutes with D_i,

$$(D_i \mathcal{T}_{\boldsymbol{q}} - \mathcal{T}_{\boldsymbol{q}} D_i)u = -\sum_{\boldsymbol{\ell}} \sum_{k=0}^{q_i} (2k+1) s_{i,1-(k \bmod 2),\boldsymbol{\ell}} P_k(\xi_i) \prod_{j \ne i} P_{\ell_j}(\xi_j). \tag{16}$$

Its squared norm follows by multiplying every squared coefficient by $\prod_j 2/(2\ell_j + 1)$, with $\ell_i = k$. There are two *face functions* per direction, not two numbers for a whole multidimensional element. The directional rank is at most $2\prod_{j \ne i}(q_j + 1)$. The software verifies both two- and three-dimensional cases. A tensor-product compatible projection also requires different scalar and derivative spaces, as in compatible discretization theory; a componentwise use of (11) in every variable is not automatically a commuting multidimensional transfer.

5.2 Why nonaffine physical maps require another term

For a one-dimensional map with positive Jacobian $J(\xi)$, write $v = J^{-1}$. A consistent physical derivative is $\Phi = vDu$. Independent physical-field projection satisfies the exact functional identity

$$vD\mathcal{T}_q u - \mathcal{T}_q(vDu) = v\, c_q(u) + [v, \mathcal{T}_q]Du, \quad [v, \mathcal{T}_q]f = v\mathcal{T}_q f - \mathcal{T}_q(vf). \tag{17}$$

The second term vanishes for constant J. It does not vanish for a general mapped element, and its norm is not determined by the two traces of the discarded potential. Moreover, interpolation of vDu on a finite source grid introduces an additional representation error if that product is not polynomial. Equation (17) describes continuous projection; a mapped nodal code must also account for that aliasing.

A counterexample is $u = P_9 - P_{11}$, $q = 8$, and $x = \xi + \alpha\xi^2/2$. The affine face-tail term vanishes identically, but projection of the rational physical derivative gives logical-measure constraint norms approximately 0.01643, 0.16258, and 0.95920 for $\alpha = 0.1, 0.3, 0.6$, respectively. Figure 5 shows this failure of an affine-only diagnostic. A practical mapped-element implementation can evaluate a candidate physical coarse state directly, or separately bound the mapping commutator. Neither shortcut is supplied by the rank-two theorem.

5.3 Connection to SpECTRE and numerical norms

The inspected SpECTRE revision is `d57433e93984c238a144f83411eda9f125759812` [6]. In `Projection.cpp`, the full-segment, non-massive projection branch forms a modal-to-nodal matrix, truncates or zero-pads modal coefficients, and applies a nodal-to-modal matrix. The pure p-projection route delegates to this branch. For a polynomial represented exactly at source nodes, this gives

$$R_{Nq} = V_q \, [I_{q+1} \; 0] V_N^{-1}, \tag{18}$$

where V_N and V_q are the source and target Legendre Vandermonde matrices. Our tests independently implement (18) for all four Gauss/Gauss–Lobatto source-target combinations. This is source-level alignment and independent reimplementation, not a linked SpECTRE run or an upstream contribution.

Once the modal coefficients are available, forming parity suffix sums costs $O(N)$ per one-dimensional line. Every candidate q then requires constant work for the injected norm and the two-mode field penalty, or $O(q)$ to assemble the complete residual including inherited coefficients. A nodal-to-modal transform still has its usual cost when coefficients are not already available. No end-to-end runtime speedup is claimed.

The theorem uses continuous norms. Gauss–Lobatto quadrature with $q + 1$ nodes does not integrate the square of a degree-q polynomial exactly. Consequently a lumped nodal norm of C_q need not equal (6). The supplied implementation uses exact modal norm weights. It does not silently substitute a quadrature norm.

6 Computational validation

All computations are deterministic and use double precision unless exact rational arithmetic is specified. Raw results, seeds, scripts, and the environment record accompany the code. The experiments address algebraic correctness, candidate screening, and subsequent evolution separately.

6.1 Independent algebra and nodal tests

Seven automated test groups verify the inherited-error identity, the prospective decision with inherited error, all four nodal quadrature combinations, endpoint and derivative projection properties, sharpness, exact rational polynomial integration, and tensor-product identities. The rational check integrates monomial representations obtained from the closed-form Legendre polynomial coefficients; it does not reuse the implemented Legendre differentiation routine. It covers the 119 degree pairs $2 \le N \le 15, 0 \le q < N$.

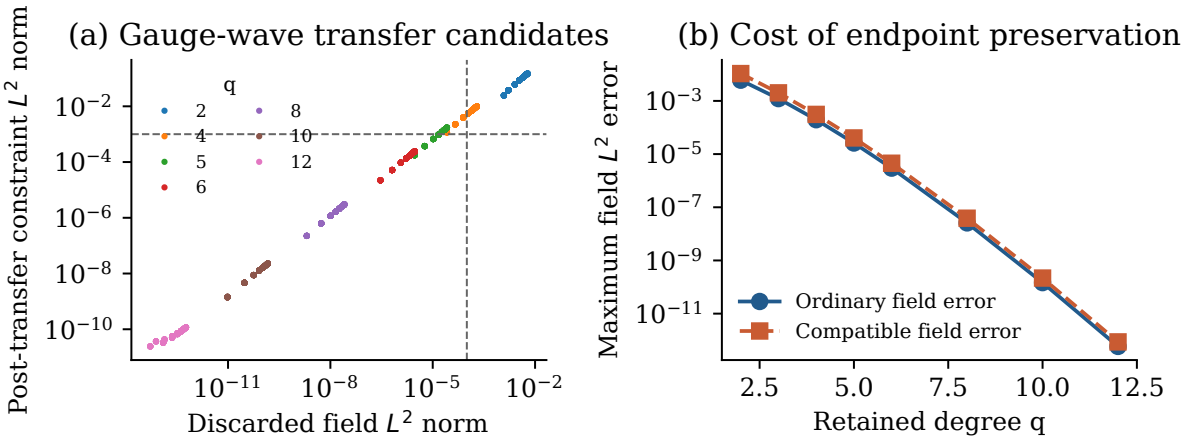


Figure 3: Gauge-wave metric-component transfers. Left: discarded field error does not alone determine the reduction constraint. Dashed lines show the illustrative tolerances $\eta = 10^{-4}$ and $\varepsilon = 10^{-3}$. Right: compatible endpoint preservation has a larger maximum field error, consistent with the exact orthogonal penalty in (13).

A separate random experiment uses $N \in \{8, 16, 24, 32, 48\}$, $q \in \{2, 4, 6\}$, and 200 coefficient vectors per pair, generated with seed 6602027 and scaled by $(n+1)^{-2}$. Direct differentiation and the two-trace formula differ by at most 4.45×10^{-16} in any modal coefficient across 3,000 cases. The computed gains are checked against weighted singular values of independently assembled differentiation and restriction matrices for 495 degree pairs (Fig. 2). These tests support implementation correctness; the mathematical guarantee comes from the proofs rather than from a finite test count.

6.2 Gauge-wave metric-component transfers

The standard gauge-wave test represents flat spacetime in nontrivial coordinates [12]. We use one metric component

$$g_{xx}(x,t) = H(x,t) = 1 - 0.1\sin(2\pi(x-t)), \qquad \Phi_{xxx} = \partial_x H. \tag{19}$$

This experiment transfers a component and its reduction variable. It does not evolve Einstein's equations or test the full harmonic gauge constraint. On each patch, H is first projected to degree $N = 24$ by 96-point Gauss quadrature, and its derivative is obtained by differentiating that polynomial. Thus the source polynomial pair satisfies the reduction constraint exactly up to arithmetic error; the analytic-to-polynomial approximation is separate from the transfer defect.

We use patch length $h = 0.5$, eight centers from -0.875 to 0.875, 32 phases $t = j/32$, and target degrees $q \in \{2, 3, 4, 5, 6, 8, 10, 12\}$, giving 2,048 transfer candidates. These are overlapping test patches and deterministic phases, not independent observational samples. Predicted and directly measured physical constraint norms differ by at most 1.12×10^{-16}. Compatible transfer produces a largest measured reduction constraint of 5.27×10^{-17} and a largest endpoint error of 2.23×10^{-16}.

Figure 3 compares field and constraint errors. To illustrate screening, impose a physical discarded-field tolerance $\eta = 10^{-4}$ and a physical constraint tolerance $\varepsilon = 10^{-3}$. The field-only test accepts 1,360 candidates. Of these, 224 have post-transfer constraint norm above ε. The joint exact test accepts the remaining 1,136. At the same constraint tolerance, field tolerances 10^{-3} and 10^{-5} accept 1,680 and 1,072 candidates, with 544 and zero constraint violations, respectively. These thresholds are illustrative choices in nondimensional units; the counts are not an estimated failure probability or a tuned optimal policy.

Table 1: Degree-14 to degree-6 transfer at $t = 0.25$. Values use the nominal step. ΔE is the immediate wave-energy change; e_ψ is the physical continuous L^2 field error.

Transfer	$\|C\|_{t_*^+}$	ΔE	$e_\psi(t_*^+)$	$e_\psi(1)$
Truncate	1.8883×10^{-4}	-1.1923×10^{-10}	1.1585×10^{-6}	6.8888×10^{-6}
Reset	1.8296×10^{-14}	$+3.0844\times10^{-5}$	1.1585×10^{-6}	6.5014×10^{-6}
Compatible	1.3150×10^{-13}	-8.3580×10^{-9}	1.7754×10^{-6}	6.6662×10^{-6}

6.3 Periodic scalar-wave evolution

To examine what an instantaneous guarantee does and does not imply, we implement an independent one-dimensional DG solver for

$$\partial_t\psi = -\Pi, \qquad \partial_t\Pi = -\partial_x\Phi, \qquad \partial_t\Phi = -\partial_x\Pi + \gamma_2(\partial_x\psi - \Phi), \quad \gamma_2 = 1. \tag{20}$$

The domain is periodic, $[-1, 1]$, with eight equal elements. A strong-form LGL nodal DG discretization uses upwind acoustic interface fluxes based on $w_\pm = \Pi \pm \Phi$, the speed-±1 characteristics of the undamped acoustic subsystem. The damping term, including its potential derivative, is evaluated in the element volume. These acoustic combinations are not the characteristic variables of the complete principal matrix when $\gamma_2 \neq 0$. The zero-speed potential receives no interface flux lift. Fourth-order Runge–Kutta integration uses nominal step $\Delta t = 0.12h/(p+1)^2$, shortened uniformly within each time interval to land exactly on the transfer and final times. Every transfer experiment is repeated with half this step.

Initial data are a right-moving smooth wave,

$$\psi(x,0) = \sin(\pi x) + 0.2\cos(3\pi x), \quad \Pi(x,0) = \Phi(x,0) = \pi\cos(\pi x) - 0.6\pi\sin(3\pi x). \tag{21}$$

The analytic solution is obtained by replacing x with $x - t$. The fine run uses degree 14 until $t_* = 0.25$, when all elements are coarsened to degree 4, 6, or 8. We compare ordinary componentwise truncation, truncation followed by $\Phi \leftarrow \partial_x\mathcal{T}_q\psi$ (reset), and (14). All continue to $t = 1$. Constraint and energy norms are integrated exactly in the modal basis; field error relative to the analytic wave is integrated at 48 independent Gauss points per element. Error is therefore measured between functions, not merely at target interpolation nodes.

The pre-transfer constraint is 3.22×10^{-13}. Ordinary coarsening injects global constraints of approximately 1.4025×10^{-2}, 1.8883×10^{-4}, and 1.3147×10^{-6} at target degrees 4, 6, and 8, matching the face-tail prediction. Reset and compatible transfer remove the immediate injection to arithmetic accuracy, up to inherited error. At degree 6, however, reset adds 3.08×10^{-5} wave energy, while compatible transfer loses 8.36×10^{-9} (Table 1). This is a measured example of energy addition by derivative resetting, not a claim that reset always adds energy.

After continued evolution, compatible transfer has a larger constraint norm than ordinary truncation in these tests. At degree 6 the final values are approximately 3.90×10^{-4} and 3.52×10^{-4}, respectively. The immediate algebraic improvement therefore does not persist as a uniform dynamical improvement. The DG interface treatment and the coarse evolution can generate fresh reduction error. All three methods converge rapidly as the target degree increases, and reset gives the lowest final field error among the three in this particular experiment. These results support screening or quantifying a transfer; they do not establish a preferred replacement evolution method.

The independent fixed-degree convergence test in Table 2 verifies the evolution implementation against the analytic solution. Halving the nominal time step in the transfer runs changes the degree-6 final field errors by less than 1.1×10^{-11} in absolute norm. The paired time-step results and all degree-4 and degree-8 results are included in the repository.

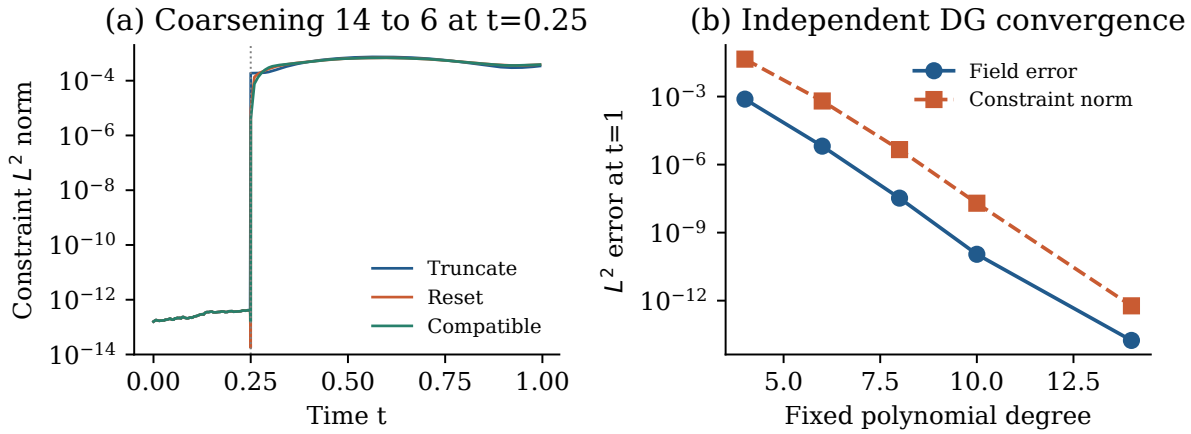


Figure 4: Left: constraint evolution for the degree-14 to degree-6 experiment, with the half-step runs shown. Compatible transfer and resetting suppress the instantaneous jump at $t = 0.25$, but later coarse evolution generates constraints again. Right: fixed-degree, no-transfer convergence against the analytic wave at $t = 1$.

Table 2: Independent fixed-degree DG convergence at $t = 1$, with $\Delta t \leq 0.06h/(p+1)^2$, without coarsening.

Degree p	Field error $\|\psi_h - \psi\|$	Constraint norm $\|C_h\|$
4	7.7048×10^{-4}	4.4701×10^{-2}
6	6.4971×10^{-6}	6.3812×10^{-4}
8	3.3188×10^{-8}	4.5829×10^{-6}
10	1.1076×10^{-10}	1.9802×10^{-8}
14	1.7610×10^{-14}	5.9054×10^{-13}

6.4 Computational assistance and verification provenance

OpenAI Codex was used to assist with topic development and literature discovery, mathematical derivation, implementation of the reference operators and tests, execution and analysis of numerical experiments, and manuscript drafting. The numerical results reported here were produced by the executable Python scripts in the public repository. Figures were generated from those results with Matplotlib; they are not generative illustrations. The algebraic checks, independent nodal calculations, singular-value comparisons and convergence checks are specified above so that a reader can audit the computations without relying on generated prose. Automated checks do not replace scientific review by the named author or establish exhaustive novelty.

7 Use in adaptation and scope of the result

A candidate coarsening can now be assessed without conflating field fidelity and reduction consistency. In the affine setting, first compute the discarded-field norm and the prospective constraint in Corollary 2. Reject the candidate if either prescribed tolerance is exceeded. If endpoint preservation is required instead, (13) gives the compatible field error exactly before committing to the alternative transfer. A decision to modify the state should remain separate from the decision to reduce degree.

This test concerns only a specified instant and specified constraints. In generalized harmonic evolution, satisfying C_{iab} does not imply satisfaction of the harmonic gauge, Hamiltonian, momentum, or all constraint-propagation relations. The physical characteristic data and symmetrizer also depend on the metric. A scalar flat-space energy argument does not certify those

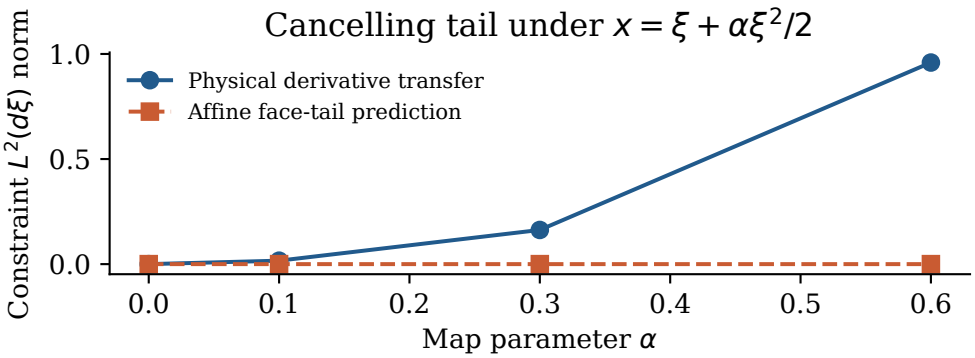


Figure 5: Nonaffine mapping counterexample. The discarded potential $P_9 - P_{11}$ has zero face-tail injection at $q = 8$, but the physical derivative projection has a nonzero constraint when the inverse Jacobian varies. Norms here use $d\xi$; 256-point Gauss quadrature evaluates the rational derivative. The affine baseline is about 5.7×10^{-12} from numerical cancellation.

structures. In particular, the metric-component test in Sec. 6 is a controlled transfer benchmark, not evidence of an accurate binary black hole waveform.

The mapped counterexample limits direct deployment: a two-trace guard alone is unsuitable for a general curved or moving mesh. A production SpECTRE contribution would need integration into its actual AMR projectors, mapped tensor variables, constraints, and boundary-data recalculation. Independent comparisons against prospective coarse-state evaluation should precede any replacement of an existing criterion. No such production integration or strong-field evolution has been performed here. The available code is a research reference implementation.

The practical opportunity is modest but concrete. Spectral simulation software can expose the predicted constraint jump and the compatible field penalty as adaptation diagnostics, including when candidate resolutions are proposed by automated tooling. Such diagnostics could support verification services for simulation workflows across scientific and engineering applications. This study does not quantify cost savings, customer demand, or patentability; its commercial value would depend on integration and validated reductions in failed or over-resolved runs.

8 Conclusion

For full-element Legendre restriction, two discarded endpoint traces determine the complete newly introduced first-order reduction constraint in one dimension. The identity supplies a sharp finite-degree amplification factor, an exact inherited-error decomposition, and a prospective constraint test. The same parity sums quantify the field penalty of a classical compatible projector and expose when a potential-only repair is impossible.

The computational evidence verifies these transfer statements while showing their limits. Field-only screening admits constraint-violating coarsenings in a deterministic metric-component test, nonaffine maps add an independent defect, and eliminating the immediate jump does not guarantee a better evolved solution. The resulting contribution is a small, explicit component for auditable spectral adaptation, with clear conditions under which it can be used and tested.

Reproducibility and availability

The repository contains the reference operators, automated tests, experiment and figure scripts, raw CSV results, environment metadata, and this manuscript's source. The code is released under the MIT license. From the repository root:

```
python -m pip install -r requirements.txt
PYTHONPATH=src python -m unittest discover -s tests -v
python experiments/run.py
python experiments/figures.py
cd paper && tectonic main.tex
```

The reported environment uses Python with NumPy 1.26.4, SciPy 1.13.1, Matplotlib 3.9.4, and pandas. No external simulation data are needed. Source references to SpECTRE are pinned to the revision identified in Sec. 5. Floating-point agreement tolerances and the distinction between continuous and nodal norms are documented in the tests. The manuscript is a research working paper and has not been peer reviewed.

Declaration of generative AI and AI-assisted technologies

During preparation of this work, OpenAI Codex was used for research assistance, code development, computational analysis and manuscript preparation, as specified in the computational provenance subsection. This disclosure describes substantive assistance beyond spelling or grammar correction. The tool is not an author. After using the tool, the author reviewed the content, approved the manuscript and disclosure, and accepts responsibility for the work.

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